

O-LEVEL PHYSICS • SYLLABUS 6091

Electromagnetism Question Set

Magnetism • Electromagnetism • Electromagnetic Induction

Topics 17–19 — 30 MCQ + 20 open-ended, with full answer key

What's inside. A complete practice set spanning the three linked topics:

- **30 multiple-choice questions** (A–D), graded easy → hard, with diagrams where needed.
- **20 open-ended questions** mixing recall, application and calculation.
- Every question tagged by topic so you can set targeted practice.
- **Full answer key** at the end: an MCQ grid with brief reasons, then worked open-ended answers with mark points.

Topic tags: 17 MAGNETISM 18 ELECTROMAGNETISM
19 INDUCTION

Written to the SEAB Singapore–Cambridge O-Level Physics syllabus 6091 (first examined 2026), Topics 17–19. Questions and figures are original teaching material, not reproductions of past papers. Uses g and standard conventions; conventional current direction throughout.

Part A — Multiple Choice (30 questions)

Choose the single best answer. One mark each. Diagrams use the usual conventions.

⊙ = current out of page ⊗ = current into page N/S = magnetic poles Field lines run N → S outside a magnet.

MAGNETISM (TOPIC 17)

1. **17** Which statement about magnetic field lines around a bar magnet is correct?
 - A They cross each other near the poles.
 - B They point from S to N outside the magnet.
 - C They are closest together where the field is strongest.
 - D They form a straight line from N to S.
2. **17** A steel bar and an iron bar are each stroked with a magnet. Compared with iron, steel is a better material for a *permanent* magnet because it...
 - A magnetises easily and demagnetises easily.
 - B is hard to magnetise but retains its magnetism.
 - C cannot be magnetised at all.
 - D is attracted only to the north pole of a magnet.
3. **17** A magnetic material is placed near the N pole of a strong magnet and becomes magnetised. The end nearest the magnet's N pole becomes a...
 - A N pole, and is repelled.
 - B N pole, and is attracted.
 - C S pole, and is repelled.
 - D S pole, and is attracted.
4. **17** Which test *conclusively* proves that a bar is a magnet (rather than just a magnetic material)?
 - A It is attracted to a known magnet.
 - B It picks up iron filings.
 - C One end of it is *repelled* by a known magnet.
 - D It is attracted to a steel nail.
5. **17** The diagram shows two bar magnets placed end to end. Which arrangement gives the field pattern with a *neutral point* (zero field) between them?



A

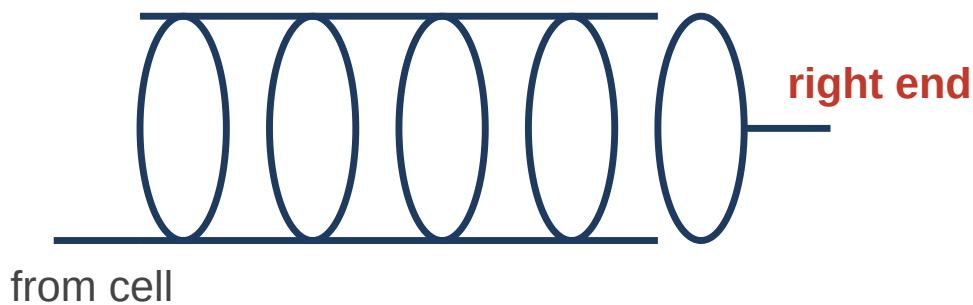


B

- A Arrangement A (N–S N–S, unlike poles facing).
- B Arrangement B (N–S S–N, like poles facing).
- C Both give a neutral point.
- D Neither gives a neutral point.

ELECTROMAGNETISM (TOPIC 18)

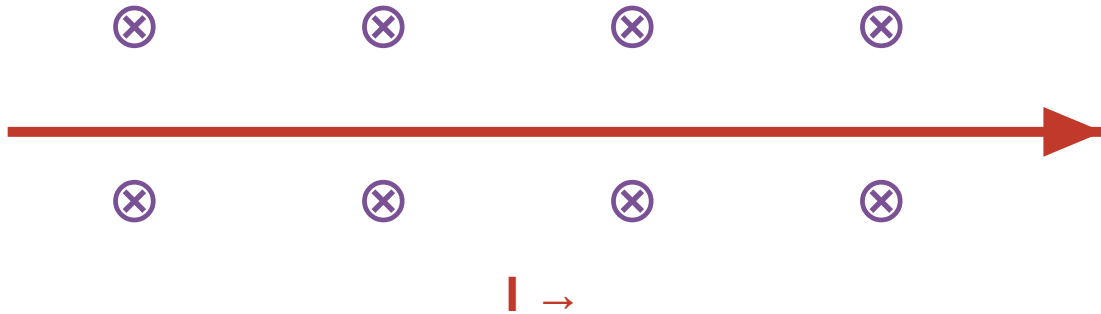
6. **18** A current flows upward through a vertical wire. Viewed from above, the magnetic field circles the wire...
- A clockwise.
 - B anticlockwise.
 - C radially outward.
 - D there is no field around a straight wire.
7. **18** The strength of the magnetic field around a long straight current-carrying wire can be increased by...
- A increasing the current.
 - B reversing the current.
 - C using a longer wire of the same current.
 - D moving the wire faster.
8. **18** A solenoid carries a current as shown, with the current flowing anticlockwise when viewed from the right-hand end. That right-hand end behaves as a...



- A N pole (use the right-hand grip / clock-face rule).
 - B S pole.
 - C neutral point.
 - D pole that reverses many times per second.
9. **18** A soft-iron core is placed inside a current-carrying solenoid. The main effect is to...
- A reverse the direction of the field.
 - B greatly strengthen the magnetic field.
 - C make the electromagnet permanent after switch-off.
 - D stop the current flowing.
10. **18** Which change would *reverse* the direction of the magnetic field of an electromagnet?
- A Increasing the number of turns.
 - B Increasing the current.
 - C Reversing the current direction.
 - D Removing the soft-iron core.

11. **18** A horizontal wire carries current from left to right through a magnetic field pointing into the page ($\otimes$). Using Fleming's left-hand rule, the force on the wire is...

B into page



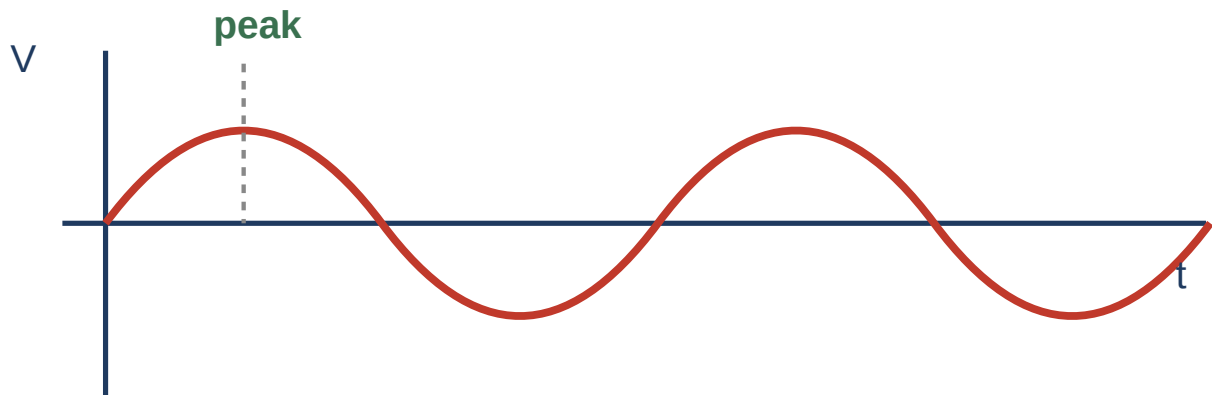
- A into the page.
B out of the page.
C vertically upward (in the plane of the page).
D vertically downward (in the plane of the page).
12. **18** In Fleming's left-hand rule, the thumb, first finger and second finger represent, respectively...
- A force, field, current.
B field, current, force.
C current, force, field.
D force, current, field.
13. **18** A current-carrying wire lies *parallel* to a magnetic field. The force on the wire is...
- A maximum.
B half its maximum value.
C zero.
D directed along the wire.
14. **18** A beam of electrons moves to the right and enters a magnetic field directed into the page. Remembering that conventional current is *opposite* to electron flow, the force on the beam is...
- A upward.
B downward.
C into the page.
D zero, because electrons are not a current.
15. **18** The turning effect (torque) on the coil of a simple d.c. motor is increased by all of the following *except*...
- A increasing the current.
B increasing the number of turns.
C using a stronger magnet.
D reversing the current every half-turn.

16. **18** The purpose of the split-ring commutator in a d.c. motor is to...
- A increase the current in the coil.
 - B reverse the current in the coil every half-turn so it keeps rotating the same way.
 - C convert a.c. into d.c. in the external circuit.
 - D reduce friction at the axle.
17. **18** A rectangular coil carrying current sits in a magnetic field. No turning effect is produced when the plane of the coil is...
- A parallel to the field.
 - B perpendicular to the field (field in the plane of the coil).
 - C at 45° to the field.
 - D a turning effect is always produced.
18. **18** A device uses an electromagnet so that a small current in one circuit switches a larger current in another. This device is a...
- A transformer.
 - B relay.
 - C fuse.
 - D rheostat.

ELECTROMAGNETIC INDUCTION (TOPIC 19)

19. **19** A bar magnet is pushed into a coil connected to a sensitive galvanometer. The meter deflects because...
- A the magnet touches the coil.
 - B the magnetic field through the coil is changing, inducing an e.m.f.
 - C the coil becomes permanently magnetised.
 - D current flows from the magnet into the coil.
20. **19** A magnet is held *stationary* inside a coil. The induced e.m.f. is...
- A at its maximum.
 - B zero, because the field is not changing.
 - C equal to the magnet's field strength.
 - D reversed compared with when it moves.
21. **19** The N pole of a magnet is pushed into a coil. By Lenz's law, the end of the coil facing the approaching magnet becomes a...
- A N pole, to oppose the approach.
 - B S pole, to attract the magnet in.
 - C neutral point.
 - D pole whose sign depends on the coil material.
22. **19** Which change would *not* increase the size of the induced e.m.f. when a magnet is moved into a coil?
- A Moving the magnet faster.
 - B Using a stronger magnet.
 - C Using a coil with more turns.
 - D Holding the magnet still, closer to the coil.

23. **19** The graph shows the output of a simple a.c. generator (rotating coil). At the instant the coil's plane is *parallel* to the field, the output voltage is...



- A zero.
B at its peak (maximum).
C half the peak value.
D negative.
24. **19** A simple a.c. generator uses slip rings. Their function is to...
- A reverse the connections every half-turn (like a commutator).
B maintain a continuous connection between the rotating coil and the external circuit.
C step up the output voltage.
D smooth the a.c. into d.c.
25. **19** An ideal transformer has 200 turns on the primary and 50 turns on the secondary. If the primary voltage is 240 V, the secondary voltage is...
- A 960 V B 120 V C 60 V D 48 V
26. **19** A transformer only works with alternating current because...
- A a.c. has a higher voltage than d.c.
B only a *changing* current produces a changing magnetic flux, which induces an e.m.f. in the secondary.
C d.c. would melt the core.
D the iron core only conducts a.c.
27. **19** An *ideal* step-up transformer raises the voltage. Compared with the primary, the secondary current is...
- A larger. B smaller. C the same. D zero.
28. **19** Electrical power is transmitted across the country at very high voltage mainly to...
- A make the current larger in the cables.
B reduce the current, so less energy is wasted as heat in the cables.
C allow thinner insulation.
D increase the resistance of the cables.
29. **19** An ideal transformer steps 240 V down to 12 V to run a lamp drawing 2.0 A. Assuming 100% efficiency, the primary current is...
- A 40 A B 2.0 A C 0.10 A D 0.010 A

30. **19** A coil rotates in a magnetic field. To *double* the peak e.m.f. of the a.c. output, the coil could be rotated...
- A half as fast, with the same field.
 - B twice as fast, with the same field.
 - C at the same speed in a weaker field.
 - D in the opposite direction.

Part B — Open-Ended (20 questions)

Show working for all calculations. Marks are shown in square brackets.

Question 1 TOPIC 17 — TESTING FOR A MAGNET

[4]

A student has two identical-looking steel bars. One is a permanent magnet; the other is unmagnetised steel.

1. Describe a simple test, using only the two bars, that would tell her which bar is the magnet. [2]
2. Explain why "one bar attracts the other" is **not** enough on its own to identify the magnet. [2]

Question 2 TOPIC 17 — TEMPORARY VS PERMANENT

[4]

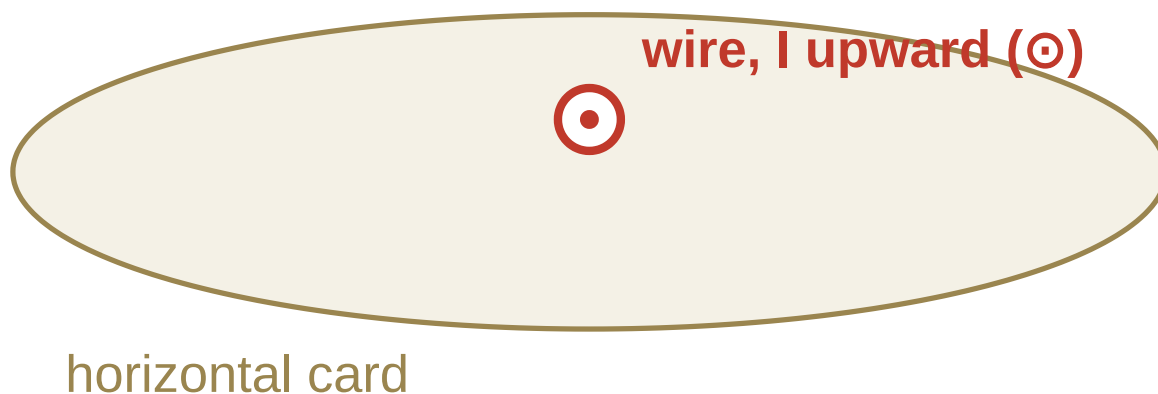
An electromagnet in a scrapyard crane uses a soft-iron core, not a steel one.

1. State one property of soft iron that makes it suitable for the core. [1]
2. Explain why the crane must use an electromagnet rather than a permanent magnet to do its job. [2]
3. Name one everyday device that uses a *permanent* magnet. [1]

Question 3 TOPIC 18 — FIELD AROUND A WIRE

[4]

A long straight vertical wire passes up through a horizontal card. A current flows upward through the wire.



1. Describe the shape of the magnetic field lines seen on the card. [1]
2. State the rule used to find the direction of the field, and give the direction (clockwise or anticlockwise) as seen from above. [2]
3. State one way to make the field pattern easier to see on the card. [1]

Question 4 TOPIC 18 — SOLENOID

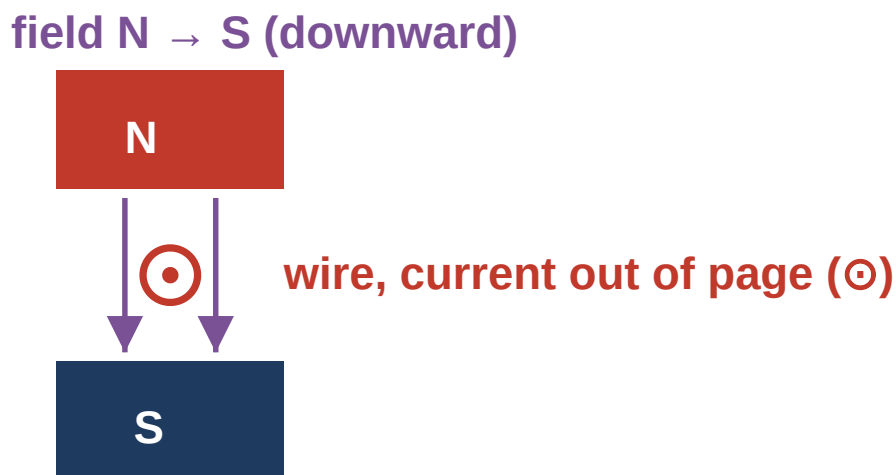
[5]

A solenoid is connected to a battery. The field it produces resembles that of a bar magnet.

1. Sketch the magnetic field pattern inside and outside a current-carrying solenoid. [2]
2. State two changes that would increase the strength of the field. [2]
3. State what happens to the poles of the solenoid if the current is reversed. [1]

Question 5 TOPIC 18 — FORCE ON A CONDUCTOR**[5]**

A straight wire is placed between the poles of a horseshoe magnet, at right angles to the field. When a current flows, the wire experiences a force.



1. State the name of the rule used to predict the direction of the force. **[1]**
2. Using the rule, state the direction of the force on the wire (left / right / up / down). **[2]**
3. State two ways the size of the force could be increased. **[2]**

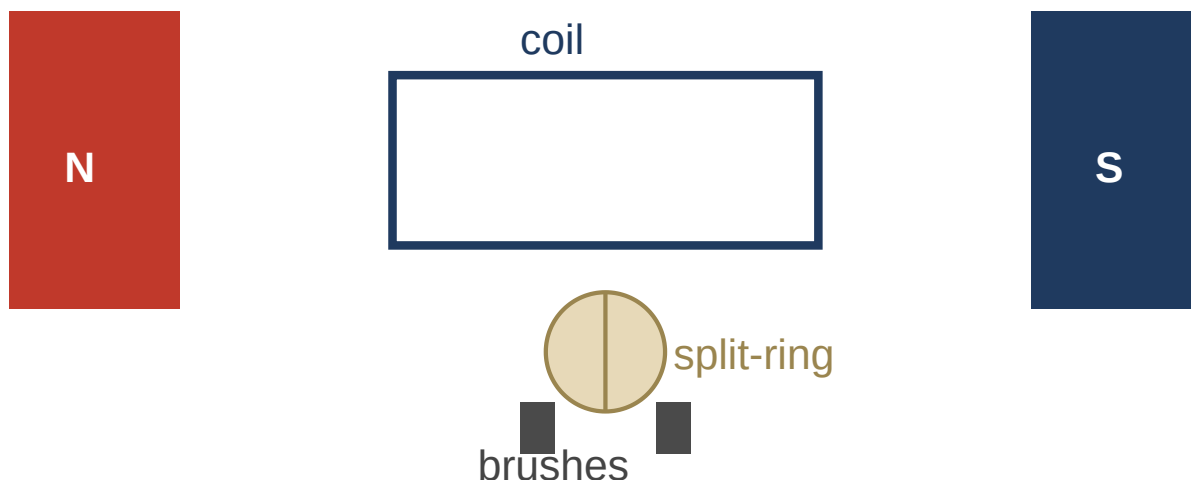
Question 6 TOPIC 18 — REVERSING THE FORCE**[3]**

In Question 5, the force pushes the wire in a certain direction.

1. State what happens to the direction of the force if *only* the current is reversed. **[1]**
2. State what happens to the direction of the force if *both* the current and the magnetic field are reversed. **[1]**
3. State what happens to the force if the wire is turned so it lies *along* the field. **[1]**

Question 7 TOPIC 18 — THE D.C. MOTOR**[6]**

The diagram shows a simple two-pole, single-coil d.c. motor.



1. Explain, in terms of forces on the two sides of the coil, why the coil turns. **[2]**
2. Explain the function of the split-ring commutator. **[2]**
3. State two changes that would make the motor turn faster. **[2]**

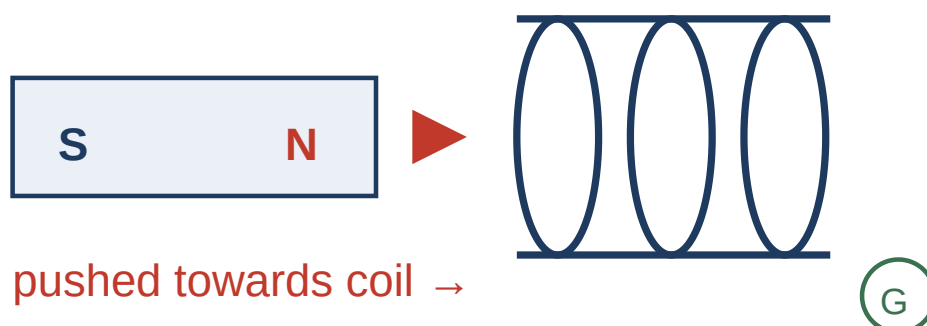
Question 8 TOPIC 19 — INDUCING AN E.M.F.**[5]**

A student moves a bar magnet in and out of a coil connected to a centre-zero galvanometer.

1. State what the galvanometer shows when the magnet is (i) pushed in, (ii) held still inside, (iii) pulled out. **[3]**
2. State two things the student could do to produce a larger deflection. **[2]**

Question 9 TOPIC 19 — LENZ'S LAW**[4]**

The N pole of a magnet is pushed towards a coil, as shown.



1. State which pole is induced on the left-hand face of the coil, and explain why using Lenz's law. **[2]**
2. Explain how this is a consequence of the conservation of energy. **[2]**

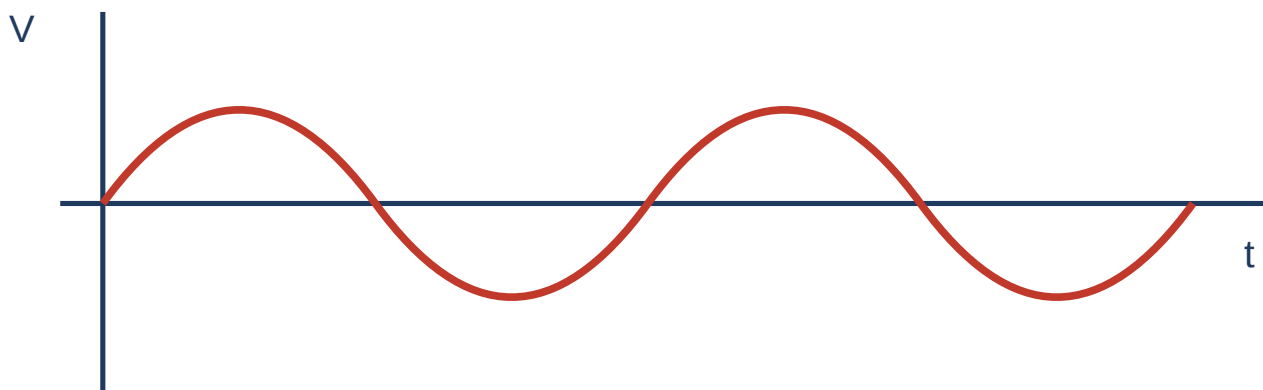
Question 10 TOPIC 19 — THE A.C. GENERATOR**[6]**

A simple a.c. generator has a coil rotating between the poles of a magnet, connected to the outside circuit by slip rings.

1. Explain why an e.m.f. is induced as the coil rotates. [2]
2. State the position of the coil at which the e.m.f. is (i) maximum, (ii) zero. [2]
3. Explain the difference between the function of slip rings (generator) and a split-ring commutator (motor). [2]

Question 11 TOPIC 19 — GENERATOR OUTPUT GRAPH**[4]**

Fig. 11.1 shows the voltage output of an a.c. generator against time.



1. On a copy of the graph, sketch the new output if the coil is rotated at **twice** the speed (describe it if you cannot draw). [2]
2. Explain the two changes to the graph you have described. [2]

Question 12 TOPIC 19 — TRANSFORMER BASICS**[4]**

An ideal transformer has a primary coil, a secondary coil, and a soft-iron core.

1. Explain how an alternating current in the primary induces a voltage in the secondary. [2]
2. State the purpose of the soft-iron core. [1]
3. Explain why a transformer does not work with a steady direct current. [1]

Question 13 TOPIC 19 — TRANSFORMER CALCULATION**[4]**

An ideal transformer is used to step down mains voltage of 240 V for a doorbell. The primary has 4800 turns and the secondary has 160 turns.

1. Calculate the secondary (output) voltage. [2]
2. State whether this is a step-up or step-down transformer, and how the turns ratio tells you. [2]

Question 14 TOPIC 19 — IDEAL TRANSFORMER POWER**[5]**

An ideal (100% efficient) transformer supplies a 12 V, 24 W lamp from a 240 V mains supply.

1. Calculate the current in the secondary coil (through the lamp). [2]
2. Using the fact that input power = output power for an ideal transformer, calculate the current drawn from the mains (primary current). [3]

Question 15 TOPIC 19 — POWER TRANSMISSION**[5]**

Electricity is transmitted from a power station to a town through cables of total resistance $4.0\ \Omega$. A power of 100 kW is transmitted.

1. If the transmission voltage is 10 kV, calculate the current in the cables. [2]
2. Calculate the power wasted as heat in the cables at this voltage. [2]
3. State and explain what happens to the wasted power if the transmission voltage is increased to 100 kV. [1]

Question 16 TOPIC 18 — THE RELAY**[4]**

A relay allows a small current in a control circuit to switch on a large current in a separate high-power circuit.

1. Explain how the relay uses an electromagnet to close the switch in the second circuit. [2]
2. State one advantage of using a relay in this way (e.g. in a car starter motor). [2]

Question 17 TOPIC 17/18 — COMPARING FIELDS**[4]**

1. State two similarities between the field of a current-carrying solenoid and that of a bar magnet. [2]
2. State two differences. [2]

Question 18 TOPIC 19 — CHARGED PARTICLE IN A FIELD**[4]**

A narrow beam of electrons travels horizontally across a vacuum tube. A magnetic field is applied vertically downward across the beam.

1. Explain why the beam is deflected. [2]
2. The current direction is taken as opposite to the electron flow. Using this, describe how you would work out the direction of deflection. [2]

Question 19 TOPIC 19 — FACTORS AFFECTING INDUCED E.M.F.**[5]**

A student investigates the factors that affect the size of the e.m.f. induced when a magnet is dropped through a coil.

1. State three factors that would increase the induced e.m.f. [3]
2. The student notices the magnet falls through the coil slightly slower than it would in free fall. Explain this observation. [2]

Question 20 TOPICS 18–19 — MOTOR VS GENERATOR**[5]**

A motor and a simple a.c. generator can look almost identical in construction.

1. State the energy transfer that takes place in (i) a motor, (ii) a generator. [2]
2. A single coil-and-magnet device is first connected to a battery, then instead connected to a bulb and spun by hand. Explain what it does in each case. [3]

Answer Key — Part A (MCQ)

Answers with a one-line reason. Accept the lettered answer alone for marking.

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	C	2	B	3	D	4	C	5	B
6	B	7	A	8	A	9	B	10	C
11	C	12	A	13	C	14	A	15	D
16	B	17	B	18	B	19	B	20	B
21	A	22	D	23	B	24	B	25	C
26	B	27	B	28	B	29	C	30	B

MCQ reasoning (the ones students trip on)

5 (B): two like poles facing (N...S S...N so the two inner poles are alike) gives opposing fields → a neutral point between them.

8 (A): right-hand grip / clock rule — anticlockwise current viewed from an end makes that end a N pole.

11 (C): field into page, current right; left-hand rule → force up the page.

14 (A): electrons move right, so conventional current is to the *left*; field into page; left-hand rule → force upward.

15 (D): reversing current every half-turn keeps rotation going but does *not* increase the turning effect — the others do.

21 (A): approaching N pole is opposed, so the near face becomes N to push back (Lenz's law).

25 (C): $V_s = V_p \times N_s/N_p = 240 \times 50/200 = 60 \text{ V}$.

27 (B): ideal transformer conserves power; if voltage goes up, current goes down ($V_p I_p = V_s I_s$).

29 (C): $V_p I_p = V_s I_s \rightarrow I_p = (12 \times 2.0)/240 = 0.10 \text{ A}$.

30 (B): peak e.m.f. rises with rotation speed, so doubling the speed doubles the peak (and also doubles the frequency).

Answer Key — Part B (Open-Ended)

Mark points tagged in red. Accept equivalent correct wording; award method marks even if the final number is wrong.

Q1 — Testing for a magnet [4]

- (a) Bring one end of a bar to *each end* of the other bar in turn [1]; a magnet will **repel** one end of the other magnet — the bar that shows repulsion is the magnet (only two magnets can repel) [1].
- (b) Attraction is not conclusive because a magnet attracts *both* a magnet and an unmagnetised magnetic material [1]; only **repulsion** proves both bars are magnets, so it uniquely identifies a magnet [1].

Q2 — Temporary vs permanent [4]

- (a) Soft iron magnetises easily and *loses* its magnetism as soon as the current is switched off [1].
- (b) The crane must be able to pick up scrap (magnet on) *and* drop it (magnet off) [1]; only an electromagnet can be switched off — a permanent magnet could never release the load [1].
- (c) Any one [1]: fridge magnet, compass needle, loudspeaker, simple d.c. motor magnets.

Q3 — Field around a straight wire [4]

- (a) Concentric circles centred on the wire, more widely spaced further out [1].
- (b) Right-hand grip rule (thumb points along the current, fingers curl in the field direction) [1]; with current upward, the field is **anticlockwise** seen from above [1].
- (c) Sprinkle iron filings on the card and tap gently (or use small plotting compasses) [1].

Q4 — Solenoid [5]

- (a) Field lines straight and parallel *inside* the solenoid (uniform, close together) [1]; emerging from one end (N), looping round the outside to the other end (S), like a bar magnet [1].
- (b) Any two [2]: increase the current; increase the number of turns per unit length; insert a soft-iron core.
- (c) The poles swap over / N and S ends are reversed [1].

Q5 — Force on a conductor [5]

- (a) Fleming's left-hand rule [1].
- (b) Field points down (N → S), current is out of the page; left-hand rule (thumb = force, first finger = field, second finger = current) → force is to the **left** [2]. (Accept a clearly reasoned direction consistent with the diagram.)
- (c) Any two [2]: increase the current; use a stronger magnet (larger field); (increase the length of wire in the field).

Q6 — Reversing the force [3]

- (a) The force reverses direction [1].
- (b) Reversing *both* current and field leaves the force in the **same** direction (two reversals cancel) [1].
- (c) The force becomes zero (no force when current is parallel to the field) [1].

Q7 — The d.c. motor [6]

- (a) The two sides of the coil carry current in opposite directions, so they feel forces in opposite directions (one up, one down) [1]; this pair of forces forms a couple that produces a turning effect, rotating the coil [1].
- (b) The commutator reverses the current in the coil every half-turn [1]; this reverses the forces at the right moment so the coil keeps rotating in the same direction instead of oscillating [1].
- (c) Any two [2]: increase the current; increase the number of turns on the coil; use a stronger magnet; (wind the coil on a soft-iron cylinder).

Q8 — Inducing an e.m.f. [5]

- (a) (i) Deflects one way as the field through the coil changes [1]; (ii) no deflection — field is not changing, so no induced e.m.f. [1]; (iii) deflects the *opposite* way as the magnet is withdrawn [1].
- (b) Any two [2]: move the magnet faster; use a stronger magnet; use a coil with more turns.

Q9 — Lenz's law [4]

- (a) The left face (facing the approaching N pole) becomes a **N pole** [1]; the induced current flows so as to oppose the change causing it, i.e. to repel the incoming magnet [1].
- (b) Work must be done pushing the magnet against this repulsion [1]; that work is the source of the electrical energy generated — energy is conserved, not created from nothing [1].

Q10 — The a.c. generator [6]

- (a) As the coil rotates, the magnetic flux/field lines through it change continuously [1]; a changing flux induces an e.m.f. across the coil [1].
- (b) (i) Maximum when the coil is *parallel* to the field (sides cutting field lines fastest) [1]; (ii) zero when the coil is *perpendicular* to the field (sides moving along the lines, not cutting them) [1].
- (c) Slip rings keep a continuous connection so the output stays as a.c. [1]; a split-ring commutator reverses the connection every half-turn to give one-directional (d.c.) output / one-way rotation [1].

Q11 — Generator output graph [4]

- (a) New curve has **double the peak height** and **half the period** (twice as many cycles in the same time) [2] (1 for amplitude, 1 for frequency).
- (b) Faster rotation cuts field lines more rapidly, so the induced e.m.f. is larger (higher peak) [1]; and one full rotation takes less time, so the frequency doubles / period halves [1].

Q12 — Transformer basics [4]

- (a) The alternating primary current produces a continuously changing magnetic flux in the core [1]; this changing flux links the secondary coil and induces an alternating e.m.f. in it [1].
- (b) The soft-iron core concentrates and channels the magnetic flux from primary to secondary (links the coils efficiently) [1].
- (c) A steady d.c. gives a constant flux — no *change* in flux means no induced e.m.f. in the secondary [1].

Q13 — Transformer calculation [4]

(a)

$$\begin{aligned}V_s/V_p &= N_s/N_p \\V_s &= 240 \times (160 / 4800) \\&= 240 \times 0.0333 \\&= 8.0 \text{ V}\end{aligned}$$

Correct formula & substitution [1]; **$V_s = 8.0 \text{ V}$** [1].

(b) Step-down [1]; because there are fewer turns on the secondary than the primary ($N_s < N_p$), so the output voltage is lower [1].

Q14 — Ideal transformer power [5]

(a)

$$\begin{aligned}P &= V_s \times I_s \\I_s &= P / V_s = 24 / 12 = 2.0 \text{ A}\end{aligned}$$

Correct method [1]; **$I_s = 2.0 \text{ A}$** [1].

(b) For an ideal transformer, input power = output power = 24 W [1]:

$$I_p = P / V_p = 24 / 240 = 0.10 \text{ A}$$

Correct method [1]; **$I_p = 0.10 \text{ A}$** [1].

Q15 — Power transmission [5]

(a)

$$\begin{aligned}P &= VI \rightarrow I = P / V \\I &= 100\,000 / 10\,000 = 10 \text{ A}\end{aligned}$$

Method [1]; **$I = 10 \text{ A}$** [1].

(b)

$$\begin{aligned}P(\text{wasted}) &= I^2 R \\&= (10)^2 \times 4.0 \\&= 100 \times 4.0 = 400 \text{ W}\end{aligned}$$

Method (uses $I^2 R$) [1]; **400 W** [1].

(c) At 100 kV the current is 10× smaller, so the wasted power ($\propto I^2$) falls by a factor of 100 (to ~4 W) — much less energy lost as heat [1].

Q16 — The relay [4]

(a) The small control current flows through a coil, making an electromagnet [1]; this attracts an iron armature, which closes the contacts of the separate high-power circuit [1].

(b) Any valid advantage [2]: a small, safe current can control a large/dangerous one; the two circuits are electrically isolated; thin control wiring can switch a heavy-current circuit (e.g. a car battery's starter motor) safely from a distance.

Q17 — Comparing fields [4]

(a) Any two similarities [2]: both have a N and a S pole; both produce the same external field shape (lines from N to S); both attract magnetic materials.

(b) Any two differences [2]: the solenoid's field exists only while current flows (a bar magnet's is permanent); the solenoid's field strength/direction can be changed by changing the current; reversing the current swaps the solenoid's poles (impossible for a bar magnet).

Q18 — Charged particle in a field [4]

(a) A moving charge is equivalent to a current; a current in a magnetic field experiences a force [1]; this force is perpendicular to both the motion and the field, so the beam is deflected [1].

(b) Take the conventional current as opposite to the electron velocity [1]; apply Fleming's left-hand rule with that current direction and the given field to get the force (deflection) direction [1].

Q19 — Factors affecting induced e.m.f. [5]

(a) Any three [3]: drop the magnet from higher / make it move faster through the coil; use a stronger magnet; use a coil with more turns.

(b) The induced current in the coil creates a magnetic field that opposes the magnet's motion (Lenz's law) [1]; this retarding force slows the fall slightly below free fall [1].

Q20 — Motor vs generator [5]

(a) (i) Motor: electrical energy → kinetic (mechanical) energy [1]; (ii) generator: kinetic (mechanical) energy → electrical energy [1].

(b) Connected to a battery, the coil carries a current in the field and experiences a turning force — it spins as a **motor** [1]. Spun by hand with a bulb connected, the rotating coil cuts field lines and induces an e.m.f., driving a current that lights the bulb — it acts as a **generator** [1]; so the same device can do both, running the energy transfer in either direction [1].

Quick teaching reminders

Left-hand rule is for the *force on a current* (motor effect); electromagnetic *induction* is about a **changing** field, so a stationary magnet gives nothing. For electron beams, flip to conventional current first. In transformers, remember power is conserved ($V_p I_p = V_s I_s$) — step up the voltage and the current steps down.

Written to the SEAB Singapore–Cambridge O-Level Physics syllabus 6091 (first examined 2026), Topics 17–19. Original teaching material, not reproductions of past papers. From 2027 the O- and N-Level examinations consolidate into the Singapore–Cambridge Secondary Education Certificate (SEC); confirm against current SEAB documents for later cohorts.